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Structure-Guided Framework for Designing Pathogen-Selective PME Inhibitors Across Plant Pathogens

Plant pathogens take advantage of pectin methylesterases (PMEs) to weaken cell walls for polygalacturonases (PGs) to cleave the impaired walls into usable saccharides, while also facilitating colonization of host tissue. These are used by a variety of pathogens, including fungi and oomycetes. Plants like the *Glycine max* (soybean) use PME inhibitors (PMEIs) to reduce the effects of invading PMEs, however this also inhibits the host's own PMEs which are essential for growth and other regulatory functions of the cell wall.

A recent study (Xia et al., 2024) demonstrates that structure-guided redesign can bias PMEI specificity toward pathogen enzymes, but this has only been shown to work mildly against a narrow set of pathogen PMEs. More broadly, the extent to which PME–PMEI interactions encode exploitable structural differences across pathogens remains unclear. Addressing this gap would enable the rational design of targeted inhibitors that selectively disrupt pathogen activity while preserving host function.

As the human population grows, pathogens rapidly evolve to overcome resistant plants, and the climate changes from human byproducts, sustainable crop protection will become increasingly important for food security. I am proposing a project to systematically model and compare PME-PMEI interfaces across a diverse set of pathogens and plants. The aim of this would be to determine the highest performing PMEIs based on different metrics, and use a select few as a basis for structure-guided redesign to make them more specific to pathogens and avoid plant PMEs.

Specific Aim 1: Systematically identify PMEIs with the strongest and most selective predicted binding profiles across plant and pathogen PMEs

We will compile a vast, representative panel of plant and pathogen (specifically oomycetes and fungi) PME and plant PMEI sequences. Using an automated workflow, I will analyze PME-PMEI using structural prediction and interface analysis to model the PME-PMEI complexes. We will use a standardized set of metrics to compare, such as high predicted pathogen binding and low predicted host binding. Once this step is finished, we will identify and rank candidate PMEI scaffolds and structural determinants of specificity.

Specific Aim 2: Use structure-guided redesign of top PMEI scaffolds to improve pathogen specificity while minimizing inhibition of plant PMEs

After selecting the most promising PMEIs and gathering data on contact patterns, we will look towards systematically modifying these PMEIs to more strongly bind pathogen PMEs and reduce bonding interactions with host PMEs, while still keeping low K_d values. If no PMEIs outperform each other significantly, I will use a representative panel. This will involve identifying the residues involved in forming the host and pathogen PME-PMEI bonds, then testing targeted substitutions predicted to reduce host interactions while preserving scaffold integrity and pathogen-facing contacts. Amino acid substitutions can be taken from proteins with similarity in structural superimposition (such as the Nicotiana tabacum Cell wall Invertase Inhibitor) can be brought in to resolve any stability/folding issues. PMEIs modified this way can be tested using AlphaFold-multimer. AlphaFold-based modeling will be used to assess fold plausibility and interface preservation. A lightweight ranking model using sequence embeddings and predicted interface features will be used in conjunction with co-immunoprecipitation and transient expression infection assays to prioritize redesign candidates.

Long-term impact

At the very least, this project would simulate and analyze a large amount of PME-PMEI data for future studies and translational applications in related contexts. If the process sees success and yields highly pathogen-specific, low K_d PMEI(s) which are tested and confirmed to work well in aiding an immune response against pathogens, further research could be done to engineer a new generation of resistant crops. It is much more likely that we will discover improved PMEIs which show a similar level of change that GmPMI1R had to GmPMI1.

Citations

Xia, Y., Sun, G., Xiao, J., He, X., Jiang, H., Zhang, Z., Zhang, Q., Li, K., Zhang, S., Shi, X., Wang, Z., Liu, L., Zhao, Y., Yang, Y., Duan, K., Ye, W., Wang, Y., Dong, S., Wang, Y., Ma, Z., & Wang, Y. (2024). *AlphaFold-guided redesign of a plant pectin methylesterase inhibitor for broad-spectrum disease resistance*. *Molecular Plant, 17*(9), 1344–1368. https://doi.org/10.1016/j.molp.2024.07.008